

Topics in Error-Correcting Codes

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Lecture 6: Interactive Coding (Cont.) (September 28, 2026)

Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Recap

Recall that last class we defined the notion of a tree code and showed how to use a tree code to construct an interactive coding scheme.

Definition 1. A tree code, T , is an online code with,

$$T : \{0,1\}^* \rightarrow \Sigma^*,$$

such that the following conditions hold:

- $|x| = |y| \implies |T(x)|_\Sigma = |T(y)|_\Sigma$.
- For every $x \in \{0,1\}^*$, $T(x)$ is the prefix of $T(x\|0)$ and $T(x\|1)$.
- There exists a constant $c \in \mathbb{N}$ such that $|\Sigma| = 2^c$.

Remark. We identify Σ with $\{0,1\}^c$. Throughout these notes, Δ and wt count individual bits, rather than output symbols. Thus, relative distance δ means that two paths that first differ at position s have encodings at bit distance at least $\delta c(t - s + 1)$ at depth t .

Last class we showed how to use a tree code to construct an interactive coding scheme.

Theorem 2 (Tree codes imply interactive coding). [1] Suppose that T is a tree-code of rate $R = 1/c$ and relative distance $\delta > 0$, where c is a positive integer. Then there is a generic deterministic transformation taking any binary protocol communicating k bits into a protocol communicating $12k/R$ bits, such that both parties recover the original transcript even if an adversary corrupts a $\delta/96$ fraction of the transmitted bits.

Schulman's Interactive Coding Scheme. At a high-level, in each round t each party $P \in \{A, B\}$ has a state $\text{st}_P^t = (a_P^t, v_P^t)$, where $a_P^t \in \{0,1,\text{Back},\text{Stay}\}$ is the move that it makes at the end of round t , and v_P^t is a vertex in the protocol tree of the original protocol Π that party P believes they are at. Initialize

$$a_P^0 = \text{Stay} \quad v_P^0 = \varepsilon.$$

Party P also maintains a history $(h_P^1, \dots, h_P^t)$ where each

$$h_P^i \in \Sigma := \{0, 1, \text{Back}, \text{Stay}\} \times \{0, 1\},$$

where

$$h_P^t = (a_P^{t-1}, b_P(v_P^{t-1})),$$

where $b_P(v_P^{t-1})$ is the bit that P would send in protocol Π if they were in round v_P^{t-1} , and if this vertex is owned by the other party Q then we define $b_P(v_P^{t-1}) = 0$. Note that $h_P^{(t)} = (h_P^1, \dots, h_P^t)$ defines a vertex in the protocol tree of Π . This vertex, denoted by $\text{Vertex}(h_P^{(t)})$, is defined by the actions in this history vector,

$$(a_P^0, \dots, a_P^{t-1}).$$

Note that

$$\text{Vertex}(h_P^{(t)}) = v_P^{t-1}.$$

The interactive coding scheme consists of $N = 2k$ rounds, where k is the number of rounds in the underlying protocol. In each round both parties transmit a message simultaneously. In round t , party P , with the other party denoted by Q , performs the following steps:

1. Append h_P^t to its history, and send the t th output block of

$$\tilde{T}(h_P^1, \dots, h_P^t).$$

where

$$\tilde{T}(h_1, \dots, h_t) = T(e(h_1), \dots, e(h_t))$$

where $e : \{0, 1, \text{Stay}, \text{Back}\} \times \{0, 1\} \rightarrow \{0, 1\}^3$ is an arbitrary bijection.

Note that each block consists of $L = 3c$ bits, and $\tilde{T}$ has relative distance $\alpha = \delta/3$.

2. Compute (a_P^t, v_P^t) from the (possibly corrupted) history blocks that it received from the other party, as follows:
 - (a) Let $z_Q^t \in \{0, 1\}^{L \cdot t}$ be the concatenation of all blocks received from Q through round t . Compute

$$\hat{h}_Q^{(t)} \in \arg \min_{h \in \Sigma^t} \Delta(\tilde{T}(h), z_Q^t),$$

breaking ties according to a fixed rule. Set

$$\hat{v}_Q = \text{vertex}(\hat{h}_Q^{(t)}), \quad \hat{b}_Q = \text{Bit}(\hat{h}_Q^{(t)}),$$

The value $\text{Bit}(\hat{h}_Q^{(t)})$ is the second component of the last symbol of the decoded history. Namely, if $\hat{h}_Q^t = (\hat{a}, \hat{b})$, then

$$\text{Bit}(\hat{h}_Q^{(t)}) = \hat{b}.$$

- (b) Choose a_P^t as follows:

- If $v_p^{t-1} = \hat{v}_Q$, then set $a_p^t = b_P(v_p^{t-1})$ if P owns this vertex, and set $a_p^t = \hat{b}_Q$ otherwise.
- If v_p^{t-1} is a proper ancestor of $\hat{v}_Q$, set $a_p^t = \text{stay}$.
- Otherwise, set $a_p^t = \text{back}$.

Then update

$$v_p^t = \text{Move}(v_p^{t-1}, a_p^t).$$

Note that the **histories are append-only**: a move back changes the protocol-tree vertex but does not erase any history symbol.

After round N , each party outputs the first k bits of its vertex, or outputs failure if its vertex has depth less than k . Any padding is then removed.

Each round communicates $2L$ bits, so the total communication is

$$M = 2LN = 6cN = 12ck. \quad (1)$$

Thus, there is a $12c$ overhead in communication complexity; in other words, the rate of the interactive coding scheme is $\frac{1}{12c} = \frac{R}{12}$.

Correctness of the Schulman's Interactive Coding Scheme

We now prove correctness. Fix the inputs and an arbitrary pattern of channel corruptions. Let $\pi^* \in \{0,1\}^N$ be the correct transcript of the extended protocol on these inputs.

For every $t \in \{0, \dots, N\}$, let w_t be the longest common prefix of v_A^t and v_B^t . Note that w_N is a prefix of π^* . This is the case since at an Alice-owned vertex, this edge is consistent with Alice's input, and at a Bob-owned vertex, it is consistent with Bob's input. We need to prove that at the end of the protocol $w_N \geq k$. We prove this via a potential function.

The potential function. For every $t \in [N]$, let $B_t \in \{0,1\}$ indicate whether at least one party incorrectly decodes the other party's history in round t . Define

$$\Phi_t = 2|w_t| - \max\{|v_A^t|, |v_B^t|\} + 4 \sum_{s=1}^t B_s.$$

In particular, $\Phi_0 = 0$.

Claim 1. For every $t \in [N]$,

$$\Phi_t \geq \Phi_{t-1} + 1.$$

Claim 2.

$$\sum_{t=1}^N B_t \leq \frac{N}{8}.$$

We first note that Claim 1 together with Claim 2 implies the correctness of the simulation. To this end, first note that Claim 1, together with the fact that $\Phi_0 = 0$ implies that $\Phi_N \geq N$. Next observe that since w_N is a prefix of both v_A^N and v_B^N ,

$$\max\{|v_A^N|, |v_B^N|\} \geq |w_N|.$$

Consequently,

$$|w_N| \geq \Phi_N - 4 \sum_{t=1}^N B_t \geq N - 4 \sum_{t=1}^N B_t \geq N/2 \quad (2)$$

where the latter inequality follows from Claim 2. This together with the fact that $N/2 = k$ implies the correctness of the simulation assuming Claim 1 and Claim 2. It thus remains to prove these two claims.

Proof of Claim 1. Fix $t \in [N]$. We consider separately the cases $B_t = 0$ and $B_t = 1$.

Case 1: $B_t = 0$. Both parties correctly decode the other party's entire history through round t . There are three possibilities.

1. The vertices coincide: $v_A^{t-1} = v_B^{t-1} = v$. The owner of v follows its prescribed bit, and the other party follows the correctly decoded announcement of that bit. Both parties therefore move to the same child of v . Hence,

$$|w_t| = |w_{t-1}| + 1$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

2. One vertex is a proper ancestor of the other. The shallower party stays and the deeper party backs up. The longest common prefix is unchanged, and the maximum depth decreases by exactly 1. Thus,

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

3. The vertices are incomparable. Both parties back up. Each party either remains in its original branch below w_{t-1} or reaches w_{t-1} itself. Their longest common prefix is therefore unchanged, while both depths decrease by 1. Hence,

$$|w_t| = |w_{t-1}|$$

and

$$\max\{|v_A^t|, |v_B^t|\} = \max\{|v_A^{t-1}|, |v_B^{t-1}|\} - 1.$$

In all three possibilities, the term $2|w_t| - \max\{|v_A^t|, |v_B^t|\}$ increases by 1. Since $B_t = 0$, the cumulative number of bad rounds does not change. Consequently,

$$\Phi_t - \Phi_{t-1} = 1.$$

Case 2: $B_t = 1$. Regardless of the decoding outcome, each party changes its vertex by at most one edge. Therefore,

$$\max\{|v_A^t|, |v_B^t|\} \leq \max\{|v_A^{t-1}|, |v_B^{t-1}|\} + 1.$$

Also, each party preserves the first $\max\{|w_{t-1}| - 1, 0\}$ bits of its path. Hence,

$$|w_t| \geq |w_{t-1}| - 1.$$

Since the cumulative number of bad rounds increases by 1, we obtain

$$\begin{aligned} \Phi_t - \Phi_{t-1} &= 2(|w_t| - |w_{t-1}|) \\ &\quad - \left(\max\{|v_A^t|, |v_B^t|\} - \max\{|v_A^{t-1}|, |v_B^{t-1}|\} \right) + 4 \\ &\geq -2 - 1 + 4 \\ &= 1. \end{aligned}$$

This proves the claim. $\square$

Proof of Claim 2. Let C denote the total number of corrupted bits. We show that

$$\sum_{t=1}^N B_t \leq \frac{2C}{\alpha L}. \quad (3)$$

where C is the total number of corrupted bits, which is at most

$$C \leq \frac{\delta}{96} \cdot M = \frac{\delta}{96} \cdot 2LN.$$

Note that Equation (3) implies that

$$\sum_{t=1}^N B_t \leq \frac{2C}{\alpha L} \leq 2 \cdot \frac{\delta}{96} \cdot 2LN \cdot \frac{1}{\alpha L} = \frac{\delta}{96} \cdot 4N \cdot \frac{3}{\delta} = \frac{N}{8},$$

as desired.

We thus focus on proving Equation (3). To this end, fix a party $P \in \{A, B\}$. Let C_P denote the number of corrupted bits transmitted by P , and let

$$\mathcal{B}_P = \left\{ t \in [N] : \widehat{h}_P^{(t)} \neq h_P^{(t)} \right\}$$

be the set of rounds in which the other party incorrectly decodes P 's history. Here,

$$h_P^{(t)} = (h_P^1, \dots, h_P^t)$$

is P 's actual history through round t . For an interval $I \subseteq [N]$, let $C_P(I)$ denote the number of corrupted bits transmitted by P in rounds belonging to I .

Fix $t \in \mathcal{B}_P$. Let i be the first position at which $h_P^{(t)}$ and $\widehat{h}_P^{(t)}$ disagree, and write

$$s = \widetilde{T}(h_P^{(t)}), \quad \widehat{s} = \widetilde{T}(\widehat{h}_P^{(t)}), \quad z = z_P^t.$$

By the definition of the decoder,

$$\Delta(\hat{s}, z) \leq \Delta(s, z).$$

Since the encoder is online, s and $\hat{s}$ agree on their first $i - 1$ output blocks. Subtracting the common contribution of these blocks gives

$$\Delta(\hat{s}_{[i:t]}, z_{[i:t]}) \leq \Delta(s_{[i:t]}, z_{[i:t]}),$$

where for every vector u , we denote by $u_{[i:t]} = (u_i, \dots, u_t)$. By the distance property of $\tilde{T}$,

$$\begin{aligned} \alpha L(t - i + 1) &\leq \Delta(s_{[i:t]}, \hat{s}_{[i:t]}) \\ &\leq \Delta(s_{[i:t]}, z_{[i:t]}) + \Delta(z_{[i:t]}, \hat{s}_{[i:t]}) \\ &\leq 2\Delta(s_{[i:t]}, z_{[i:t]}) \\ &= 2C_P([i, t]). \end{aligned}$$

Thus every $t \in \mathcal{B}_P$ has a witness interval $I_t = [i, t]$ satisfying

$$C_P(I_t) \geq \frac{\alpha L}{2} |I_t|. \quad (4)$$

We select disjoint witness intervals that cover $\mathcal{B}_P$ as follows. Initially, all times in $\mathcal{B}_P$ are unmarked. Choose the largest unmarked time t , select its witness interval $I_t = [i, t]$, and mark every time in $\mathcal{B}_P \cap I_t$. Repeat until all times in $\mathcal{B}_P$ are marked.

After selecting $I_t = [i, t]$, every remaining unmarked time is strictly smaller than i . Hence every subsequently selected interval ends before i . It follows that the selected intervals are pairwise disjoint. Let $\mathcal{I}_P$ denote this collection. Since these intervals cover $\mathcal{B}_P$, we have

$$|\mathcal{B}_P| \leq \sum_{I \in \mathcal{I}_P} |I| \leq \frac{2}{\alpha L} \sum_{I \in \mathcal{I}_P} C_P(I) \leq \frac{2C_P}{\alpha L},$$

where the second inequality follows from (4), and the last inequality uses disjointness.

Since a round is bad if at least one history is decoded incorrectly, and $C = C_A + C_B$, we conclude that

$$\sum_{t=1}^N B_t = |\mathcal{B}_A \cup \mathcal{B}_B| \leq |\mathcal{B}_A| + |\mathcal{B}_B| \leq \frac{2C}{\alpha L}.$$

This proves (3). □

Construction of Tree Codes

We still need to argue that tree codes exist. At first, it is not immediately clear that such an object exists. For example, a natural first attempt is to label every edge of the complete binary tree of depth N independently and uniformly with a string in $\mathbb{F}_2^c$. The encoding of an input is the concatenation of the edge labels on its root-to-vertex path.

Random labeling does not give a tree code. It turns out that random labeling does not give a good tree code.¹ To see why, consider any internal vertex v . Its two outgoing edges have equal labels with probability 2^{-c} . If these labels agree, then this violates the definition of a tree code (for every $\delta > 0$).

There are $2^N - 1$ internal vertices, and the collision events at different vertices are independent. Consequently,

$$\Pr[T \text{ is a tree code}] \leq (1 - 2^{-c})^{2^N - 1} = \text{negl}(N).$$

The issue is that there are exponentially many independent opportunities for a short obstruction to occur. As opposed to an error correcting code that only needs to create distance over the full codeword, a tree code must create distance for every separate paths that diverged only one step ago.

Would making sibling labels distinct fix the problem? No. Suppose that each vertex independently receives a uniformly random ordered pair of distinct labels. Fix an integer $\ell > 1/\delta$. In a depth- ℓ subtree, compare the paths 0^ℓ and $10^{\ell-1}$. Their first labels differ, but their remaining $\ell - 1$ pairs of labels can all agree. This event has probability

$$2^{-c(\ell-1)} > 0.$$

On this event the bit distance is at most c , whereas the required distance is $\delta c \ell > c$. At depth $N - \ell$, there are $2^{N-\ell}$ disjoint subtrees in which this obstruction can occur independently. Thus this modified random labeling also fails with probability tending to one.

Lesson: Fixing one particular short obstruction does not address the repeated independent opportunities for a slightly longer obstruction.

Idea: Use Linearity

For a linear code, the distance between two codewords becomes the weight of a single codeword:

$$\Delta(T(x), T(x')) = \text{wt}(T(x \oplus x')).$$

Since T is required to be an online encoding, we must use a lower triangular matrix for the encoding. It turns out that if we take a random lower triangular matrix that is **Toeplitz**, meaning the coefficients repeat along diagonals, then the resulting encoding is **time invariant** in the following sense: For every vector $z \in \{0, 1\}^t$ such that $z_s = 1$ and $z_1 = \dots = z_{s-1} = 0$, it holds that

$$T(z) = T(0^{s-1} \| z_s, \dots, z_t) = 0^{c \cdot (s-1)} \| T(z_s \dots, z_t), \quad (5)$$

where we think of $T(z_s, \dots, z_t)$ as a bit string in $\{0, 1\}^{c \cdot (t-s+1)}$.

As we show below a random linear code which is time invariant, in the sense of Equation (5), is a good tree code with high probability (for a sufficiently large constant c , which depends on the desired distance δ).

¹ This is in contrast to the fact that a random error correcting code is a good code with high probability.

The Linear Construction

Choose coefficient blocks uniformly at random

$$a_0, a_1, \dots, a_{N-1} \leftarrow \mathbb{F}_2^c.$$

At time t , output

$$y_t = \sum_{j=1}^t a_{t-j} x_j = a_0 x_t + a_1 x_{t-1} + \dots + a_{t-1} x_1. \quad (6)$$

All operations are over $\mathbb{F}_2$: multiplication by a bit either keeps or zeros a block, and addition is bitwise XOR. In particular,

$$\begin{aligned} y_1 &= a_0 x_1, \\ y_2 &= a_1 x_1 + a_0 x_2, \\ y_3 &= a_2 x_1 + a_1 x_2 + a_0 x_3. \end{aligned}$$

Equivalently,

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} a_0 & 0 & 0 & \dots & 0 \\ a_1 & a_0 & 0 & \dots & 0 \\ a_2 & a_1 & a_0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{N-1} & a_{N-2} & a_{N-3} & \dots & a_0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_N \end{pmatrix}.$$

The matrix is lower triangular, which gives online encoding. In addition, the matrix is Toeplitz, meaning that it is constant on each diagonal. This implies that the construction is time invariant, i.e., for every $z \in \{0, 1\}^t$ such that $z_s = 1$ and $z_1 = \dots = z_{s-1} = 0$,

$$T(z) = 0^{c \cdot (s-1)} \| T(z_s, \dots, z_t).$$

Thus, the difference between the labels of two siblings is always a_0 , regardless of the vertex or its depth. Thus, satisfying $\text{wt}(a_0) \geq \delta c$ takes care of every sibling pair simultaneously. The same phenomenon holds for longer divergences. For example, every pair whose divergent suffix has length two is governed by just one of

$$T(1, 0) = (a_0, a_1) \quad T(1, 1) = (a_0, a_0 + a_1)$$

If both of these vectors have Hamming weight at least $\delta \cdot 2c$ then this takes care of all such pairs everywhere in the tree. More generally, the following claim holds:

Claim 3 (Only inputs starting with 1 need to be checked). The encoder in (6) is a tree code of relative distance δ if and only if, for every $1 \leq \ell \leq N$ and every $z \in \mathbb{F}_2^\ell$ with $z_1 = 1$,

$$\text{wt}(T(z)) \geq \delta c \ell. \quad (7)$$

Proof. Suppose (7) holds. Fix any $t \in [N]$ and any two paths $x, x' \in \mathbb{F}_2^t$. Denote by $s \in [t]$ the first point in which these two vectors diverge; i.e., $x_s \neq x'_s$ and $x_i = x'_i$ for every $i \in [s-1]$. Let

$$\ell = t - s + 1, \quad z = (x_s + x'_s, \dots, x_t + x'_t).$$

Then $z_1 = 1$ and $x + x' = 0^{s-1} \| z$. By linearity and time-invariance,

$$\begin{aligned} T(x) + T(x') &= T(x + x') \\ &= T(0^{s-1} \| z) \\ &= 0^{c \cdot (s-1)} \| T(z). \end{aligned}$$

Therefore

$$\Delta(T(x), T(x')) = \text{wt}(T(z)) \geq \delta c \ell = \delta c(t - s + 1).$$

Conversely, apply the tree-code condition to the pair $z, 0^\ell$, whose first difference is at position 1. $\square$

This is the main conceptual step. Linearity removes the common history; time-invariance removes the starting time. At length ℓ , only the $2^{\ell-1}$ difference patterns beginning with 1 remain. There is no additional union bound over the vertices of the tree or over the times at which divergence begins.

A random coefficient sequence works

Let

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

be the binary entropy function.

Theorem 3 (Linear tree-code existence). *Fix $0 < \delta < 1/2$, and choose an integer c such that*

$$c(1 - H(\delta)) \geq 2. \tag{8}$$

For every depth N , if $a_0, \dots, a_{N-1}$ are independent and uniform in $\mathbb{F}_2^c$, then the encoder in (6) is a tree code of relative bit distance at least δ and rate $1/c$ with probability greater than $1/2$.

Proof. Fix any $N \in \mathbb{N}$. To prove Theorem 3 we need to prove that

$$\Pr[\forall t \in [N] \ \forall z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) \geq \delta c t] \geq \frac{1}{2}$$

By the union bound

$$\begin{aligned} &\Pr[\forall t \in [N] \ \forall z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) \geq \delta c t] = \\ &1 - \Pr[\exists t \in [N] \ \exists z \in \{0, 1\}^{t-1} : \text{wt}(T(1 \| z)) < \delta c t] \geq \\ &1 - \sum_{t=1}^N 2^{t-1} \Pr[\text{wt}(T(1 \| z)) < \delta c t] \end{aligned}$$

Note that it suffices to prove that for every $t \in [N]$ and every $z \in \{0, 1\}^{t-1}$

$$\Pr[\text{wt}(T(1\|z)) < \delta ct] \leq 2^{-2t}. \quad (9)$$

This is the case since Equation (9) implies that

$$\sum_{t=1}^N 2^{t-1} \Pr[\text{wt}(T(1\|z)) < \delta ct] \leq \sum_{t=1}^N 2^{t-1} \cdot 2^{-2t} \leq \sum_{t=1}^N 2^{-(t+1)} \leq \frac{1}{2},$$

as desired. Thus, it remains to prove Equation (9). To this end note that $T(1\|z)$ is a truly uniform vector in $(\mathbb{F}_2^c)^t$. Thus

$$\begin{aligned} \Pr[\text{wt}(T(1\|z)) < \delta c\ell] &= 2^{-c \cdot t} \sum_{\substack{r \in \mathbb{Z} \\ 0 \leq r < \delta \cdot c \cdot t}} \binom{c \cdot t}{r} \\ &\leq 2^{-c \cdot t} \cdot 2^{c \cdot t \cdot H(\delta)} \\ &= 2^{-c \cdot t \cdot (1 - H(\delta))} \\ &\leq 2^{-2t}, \end{aligned}$$

where the second inequality follows from the entropy bound assuming $\delta < \frac{1}{2}$, and the last inequality follows from the definition of c (see Equation (8)).

□

References

- [1] Leonard J. Schulman. Deterministic coding for interactive communication. In *Proceedings of the 25th Annual ACM Symposium on Theory of Computing (STOC)*, pages 747–756. ACM, 1993.